Acharya Shanti Sagar Foundation-Manuscripts Conservation and Restoration Project Closing Report- Shri Mahaveer Bramhacharyashram Karanja Lad





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1. **Title of the Project:** Conservation and Restoration Jinwani, Karanja Lad
2. **Location:** Shri Mahaveer Bramhacharyashram, Jain Gurukul, Karanja Lad, Maharashtra
3. **Period:** October 2024 to 9th August 2025
4. **Type of Objects:** Palm Leaf, Paper Manuscript, Archival Books, Post card, Handwritten Letter
5. **No. of Manuscripts:** 221
6. **No. of folios:** 28268

Type of Manuscript	No. of Manuscript	No. of Folios
Palm Leaf	4	60
Paper Manuscript	189	20524
Archival Book	28	7684
Total	**221**	**28268**

7. **New Added:**

Type of Manuscript	No. Of Manuscript	No. of Folios
Paper Manuscript	**1**	366
Handwritten Letter	-	500 Approx
Total		**866**

After the completion of the conservation work at Shri Mahaveer Brahmacharyashram, Karanja Lad, they added one manuscript containing 366 folios and a handwritten document of approximately 500 folios. Therefore, a total of 866 folios is currently pending conservation from Shri Mahaveer Brahmacharyashram, Karanja Lad.

8. **Prepared by:** Kartick Kayal Lead Conservator and Reporting Officer, Acharya Shanti Sagar Foundation, Bangalore.
9. **Date of Submission:**

10. Introduction

The conservation and restoration project at Shri Mahaveer Brahmacharyashram, carried out by the Acharya Shanti Sagar Foundation, was undertaken to stabilize and preserve a valuable collection of manuscripts and archival materials of significant historical and cultural importance.



The collection comprises approximately 12th to 18th century palm-leaf manuscripts and 15th to 20th century paper manuscripts, along with several archival books.

These manuscripts represent a wide range of writing materials, including different types of paper and ink. The collection features various scripts such as Hale Kannada, Devanagari, and Modi, and encompasses multiple languages including Kannada, Sanskrit, and Marathi.



Over the centuries, the manuscripts have suffered from diverse forms of deterioration, primarily due to insect infestation, environmental fluctuations, exposure to light, improper handling, faulty past restoration practices, and inadequate storage conditions.

Recognizing the immense cultural and historical value of this collection, the project aimed to stabilize the condition of the manuscripts, prevent further deterioration, and ensure their long-term preservation. This initiative also seeks to safeguard this invaluable heritage for future generations, enabling continued research, education, and cultural understanding.

11. Objective



The objective of conserving old manuscripts is to preserve their physical integrity, historical authenticity, and cultural significance by preventing further deterioration and ensuring their long-term stability. Conservation aims to protect fragile manuscripts from environmental damage, biological threats, and physical wear, while maintaining their original script, materials, and artistic elements. It also ensures that these valuable records of knowledge and heritage remain accessible for study, research, and future generations, without compromising their authenticity.

12. Scope of Work

The scope of the conservation and restoration project at Shri Mahaveer Brahmacharyashram included the following major tasks:

- Preliminary survey of all palm-leaf and paper manuscripts, archival books, and related materials.
- Pagination and notes.
- Preparation of item-wise condition reports with photographs and technical notes.

- Cataloguing of manuscripts by script, language, material type, and period.
- The treatment shall be carried out according to the specific condition and conservation needs of each manuscript.
- Guidance to local custodians on handling, storage, and preventive conservation practices.
- Documentation of the entire process for institutional records and future reference.

13. Methodology

The conservation and restoration process was carried out in a systematic Syntenic and ethical manner, following internationally accepted conservation standards. The methodology involved the following steps:

13.1. Documentation and Initial Assessment

- Each manuscript and archival book was high regulation photographed and assigned a unique identification number.
- A preliminary condition report was prepared, noting the Title, size, type of material, script, language, physical condition, and type of deterioration.
- The identification, measurement, and documentation of specific damages (such as tears, holes, stains, insect activity, and previous repairs) were recorded.

13.2. Scientific Testing

- pH testing of paper to determine acidity.
- Solubility tests on ink and pigments to ensure safe application of consolidates or solvents.
- Observation under magnification to examine fibre structure, fungal growth, or insect damage.



13.3. Dry Cleaning or Surface Cleaning



- **Palm-leaf manuscripts:** Cleaned with soft brushes, cotton swabs, and rubber dust to remove surface dust and organic debris.
- **Paper manuscripts:** Dry cleaning done using soft brushes, microfiber cloths, and archival erasers to remove loose dirt without damaging ink or writing.
- Care was taken to avoid moisture or abrasion on fragile areas.

13.4. Mechanical Cleaning

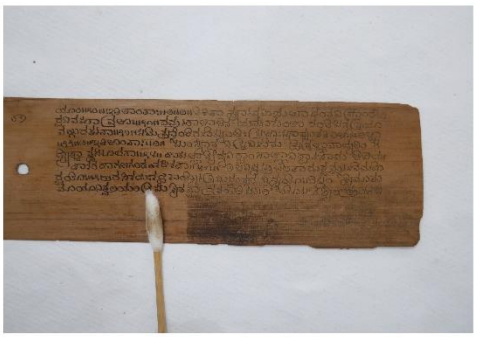


Localized Dirt Removal- Persistent dirt or stains are carefully removed using archival erasers or rubber crumbs. Movement is always light and in one direction—never circular, to prevent abrasion of ink or fibres.

Removal of Solid Particles- Small insect remains, cobwebs, or mud deposits are removed using Conservation grads tools like fine tweezers, surgical knife or Many types of micro spatula.

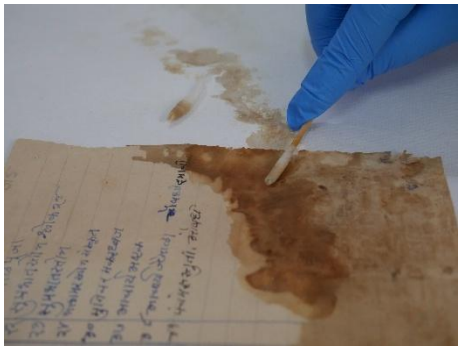
13.5. Solvent Cleaning

- **Localized Stain Minimize-** Many types of Stubborn stains, acidic deposits, or fungal marks that cannot be removed through mechanical cleaning are treated using appropriate solvent-based or aqueous solutions. The cleaning solution is applied gently with a cotton swab or soft brush, always in controlled amounts to avoid ink smudging or fibre damage.



- **Adhesive and Residue Removal-** Old adhesive tape, glue residues, or oily deposits are carefully removed using suitable solvents after conducting solubility tests to ensure that the ink, pigments, or paper are not affected.

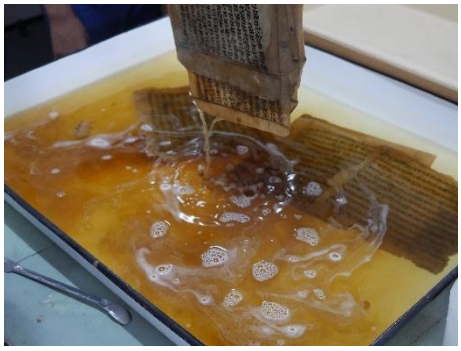




- **Testing and Safety Measures-** Before application, spot testing is performed on a small and inconspicuous area to check ink stability and material compatibility. All chemical cleaning is carried out under proper ventilation, wearing gloves and using archival-grade, reversible materials.

13.6. De-Acidification (Aqueous and Non-Aqueous)

De-acidification is a chemical treatment used to neutralize acidic components in paper or manuscripts and to deposit an alkaline buffer to prevent future acid deterioration. Acidic paper becomes brittle, discoloured, and fragile over time; therefore, de-acidification is an essential step in long-term preservation.



- **Aqueous De-Acidification Method**- This method uses an alkaline solution dissolved in water to neutralize acids in the paper. It is effective but can only be used on documents where inks and pigments are insoluble in water.
- **Non-Aqueous De-Acidification-** This method uses alcohol- or solvent-based alkaline solutions apply with brush and spray and does not involve direct contact with water—making it safer for sensitive manuscripts.

Precautions for Both Methods

- Conduct pH, ink stability, and solubility tests before treatment
- Use only archival-grade, reversible, and stable materials
- Monitor temperature, humidity, and drying process
- Avoid over-alkalization, which can make paper brittle in the long term

13.7. Joining Filling and Mending



Joining Filling and Mending refers to the localized repair of damaged areas such as fragile tears, breaks, holes, or missing portions of the Palm leaf manuscript or paper manuscript. Using Conservation grad adhesive, Japanese Kozo, Thai Kozo paper and tissue paper used to restore structural strength, prevent further tearing, and retain original appearance with minimal visual intervention.

13.8. Full lining

Full lining is a more extensive conservation technique in which the entire backside (or sometimes both sides) of a fragile manuscript or page is reinforced with a thin support layer. It is typically used when the paper or leaf is extremely brittle, weak, fragmented,

unable to support its own weight, or has multiple tears, losses, or insect damage beyond local repair.



In such cases, full lining is carried out using conservation-grade adhesive with Japanese or Thai Kozo buffered tissue paper. This process provides full support to weak or fragmented paper, increases flexibility and durability, and stabilizes heavily damaged manuscripts for safe handling and long-term storage.

13.9. Flattening

Flattening is a controlled process used to reduce or remove creases, wetness, folds, warping, or distortions in paper or manuscript leaves. After all treatment the manuscript is gently placed indite the felt or between blotting paper and under light pressure to restore its original planar form without causing stress to the fibres. This process improves the manuscript's structural stability, visual appearance, and suitability for safe storage or further treatment. Care is taken to ensure that inks, pigments, or fragile areas are not disturbed during the procedure.



13.10. Providing Cover



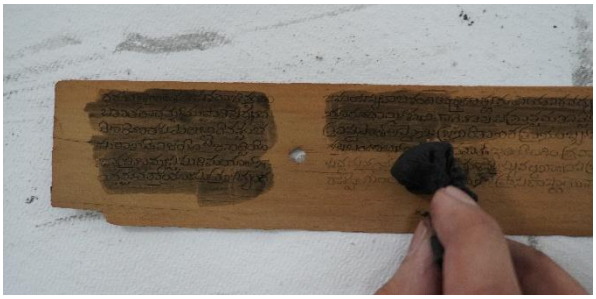
Providing a cover is the final stage of conservation, aimed at ensuring the long-term protection of the restored manuscript. After all conservation treatments are completed, the manuscript is securely placed in Conservation grade mound board or custom-made, archival-quality protective cover or enclosure. To protect the manuscript from dust, light, humidity fluctuations, physical damage, and handling stress.



13.11. For palm Leaf

Re-inking and re-oiling are also done along with the above treatment.

- **Re-inking** is the process of restoring faded writing on palm leaf manuscripts. The original engraved letters are gently cleaned, and a natural carbon-based ink is applied over the text. After the ink settles into the grooves, the excess is carefully wiped off, so only the letters remain dark and visible again. It helps make the script readable without damaging the original inscriptions.



- **Re-oiling** is a conservation process used to restore flexibility, prevent brittleness, rehydrate and protect palm leaf manuscripts from insects and environmental damage.



13.12. Storage Recommendations

Proper storage is essential to ensure the long-term preservation of conserved manuscripts after treatment. Safe storage creates a stable environment that prevents further deterioration caused by environmental conditions, handling, or biological activity. We suggest owner of manuscript **Controlled Environment-** Temperature should be maintained around **18–22°**C. Relative humidity (RH) should remain between **45–55%**, avoiding fluctuations that can cause warping, brittleness, or Mold growth. Manuscripts are housed in acid-free, lignin-free archival boxes, folders, or envelopes. Keep insect repellent materials inside the showcase. Store in a dark, clean, well-ventilated cabinet or storage room and regular monitoring.

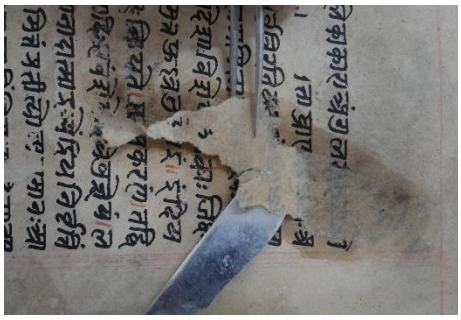
14. Challenges Faced During the Project

During the conservation and restoration process, several challenges were encountered that required careful planning, technical expertise, and adaptive solutions.

14.1. Condition of Manuscripts

- Severe brittleness, fragmentation, and powdering of old paper manuscripts made handling extremely difficult. In some manuscripts treated with non-aqueous methods, the folios have stuck together, making separation very difficult and challenging.





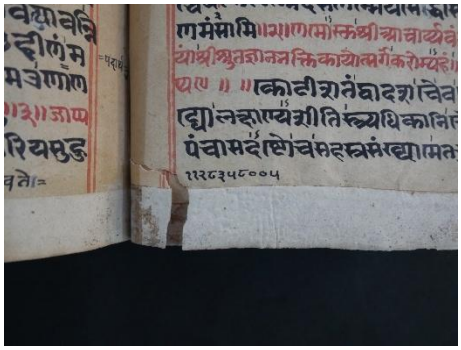
- Advanced insect damage, fungus stains, and previous faulty repairs had caused structural weakness those case treatment is challenging.

14.2. Variety of Materials

- Manuscripts were written on different types of materials such as two types of palm leaves, handmade papers, and modern paper or machine-made paper. Different types of folios, folios thickness, different types of fibbers, sizing, and different types of deterioration etc.



- Multiple scripts (Devanagari, Modi, Hale Kannada, etc.) and languages (Sanskrit, Marathi, Kannada) required accurate identification and cautious treatment.
- Different inks—such as carbon-based, turmeric, sindoor, cinnabar, mineral-based, iron gall, and other organic dyes—reacted differently to moisture and solvents. Each type of pigment or ink responded uniquely to the treatment, which made the process challenging.



- Different binder - Binders are substances used to hold pigments or particles together and help them adhere to the writing or painting surface. Different binders have different levels of longevity over time. Used in traditional manuscripts, paintings, and inks- Gum Arabic, Starch Paste, Tragacanth Gum, Plant Resins, Vegetable Oils etc. For those reasons difficult or challenging to conservation process.

14.3. Handling Fragile Items

Handling fragile manuscripts is challenging because the paper or palm-leaf sheets are often brittle, torn, powdered, or fragmented due to age, acidity, humidity, temperature, heavy light, insects, and past damage. Even slight pressure, movement, or moisture can cause further breakage or loss of text. Ink and pigments may flake off or react to touch or treatment.



14.4. Chemical Treatment Limitations

- De-acidification (aqueous and non-aqueous) is challenging in paper manuscripts some time the paper is often very fragile, and the inks or pigments used may react or get damaged by the chemicals or moisture in the process. Every manuscript has different materials and conditions, so one method cannot be safely applied to all, making the process delicate and complex.

15. Overcoming the Challenges

- Despite the extremely fragile condition of the manuscripts, the conservation team successfully addressed the challenges through careful planning, expert guidance, teamwork, and the use of appropriate conservation materials.
- The process began with detailed discussions and planning by senior conservator from another museum and conservation laboratory. As well as research and visit many conservations related web site. During this phase, the condition of the manuscripts, the types of challenges, and suitable treatment techniques were carefully evaluated. The team members, with support from knowledgeable friends and colleagues, contributed their skills and suggestions to develop practical and safe conservation strategies.
- To achieve the best results, updated conservation-grade materials, archival adhesives, and tested chemicals compatible with historical manuscripts were used to ensure long-term stability. Every step of the treatment was carried out slowly and systematically to avoid any further damage.
- Through this collaborative effort, technical expertise, and adherence to proper conservation methodologies, the challenges were effectively overcome, and the manuscripts were successfully stabilized and preserved for future generations.

16. Results After Treatment

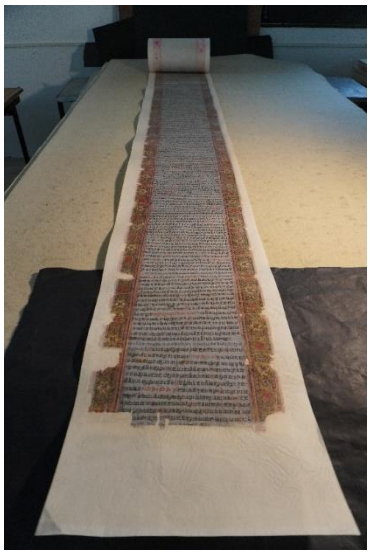
After conservation and restoration, the manuscripts are now in a stable, clean, and well-preserved condition. The structural strength of fragile folios has been restored, surface dirt and insect damage have been safely removed, and brittle or torn pages have been

repaired using archival materials. Acidic papers were neutralized, warped leaves were flattened, and each manuscript has been properly rehoused in protective, acid-free storage.

The manuscripts are now safer to handle, more readable, and protected for long-term preservation. Their historical, cultural, and research value has been successfully safeguarded for future generations.













17. Conservation Team Details

- **Project Coordinator:** Mrs. Alka Mehta
- **Lead Conservator & Reporting officer:** Mr. Kartick Kayal
- **Conservator:** Mr. Sumanta Sekhar Gayen
- **Conservator:** Mr. Dipanjan Modak
- **Conservation executive:** Mr. Mrinmoy Samadder
- **Intern:** Aniruddha Chakraborty, Samrat Ghosh, Shubham Pratap Singh Baghel, Kadambari Bhimte, Arpan Mondal, Supriya Roy, Alok Yadav,

18. Conservation Team Photo



Thank you